Lecture 9

Complex derivatives and Cauchy's theorem

Objectives. *Complex differentiability requires the same derivative in every direction. We prove that a holomorphic function on a star-shaped domain has a primitive. Its integral around every closed curve in that domain is zero. This result supplies the contour arguments used in the next lectures.*

By the end of this lecture you will be able to:

- Test complex differentiability using the real differential.
- Differentiate a power series and the principal logarithm on their domains.
- Prove Goursat's theorem and construct a primitive on a star-shaped domain.
- Check when a contour argument can use that construction.

9.1 The complex derivative

Let $\Omega \subset \mathbb{C}$ be open. Write $z = x + \mathrm{i}y$ and $f(z) = u(x, y) + \mathrm{i}v(x, y)$. The symbol i denotes the imaginary unit.

Definition 9.1. The function $f : \Omega \to \mathbb{C}$ is *complex-differentiable* at $a \in \Omega$ if the limit

$$f'(a) = \lim_{\substack{h \to 0 \\ h \in \mathbb{C} \setminus \{0\}}} \frac{f(a+h) - f(a)}{h}$$

exists. The function is *holomorphic* on Ω if it is complex-differentiable at every point of Ω.

The increment h approaches zero through all complex directions. For example, the function $f(z) = z^2$ satisfies

$$\frac{(a+h)^2 - a^2}{h} = 2a + h \longrightarrow 2a.$$

Complex conjugation behaves differently. For $f(z) = \overline{z}$, the quotient at any point is $\overline{h}/h$. It is 1 along the real axis and -1 along the imaginary axis. Hence conjugation has no complex derivative, although its real partial derivatives exist everywhere.

Recall the real differential from multivariable calculus, identifying $h \in \mathbb{C}$ with a vector of $\mathbb{R}^2$. Real differentiability at a means $f(a+h) = f(a) + Df(a)h + o(|h|)$, where $Df(a)$ is a real-linear map on $\mathbb{R}^2$.

Proposition 9.2 (Cauchy–Riemann equations)**.** *The function f is complex-differentiable at a if and only if it is real-differentiable there and its partial derivatives satisfy*

$$u_x = v_y, \qquad u_y = -v_x.$$

In this case $f'(a) = u_x + \mathrm{i}v_x$.

Proof. A complex derivative gives the expansion $f(a+h) = f(a)+f'(a)h+o(|h|)$. Multiplication by $f'(a) = A + \mathrm{i}B$ has the real matrix

$$Df(a) = \begin{pmatrix} A & -B \\ B & A \end{pmatrix}.$$

The expansion proves real differentiability with this differential. Its columns give the stated partial derivatives. Conversely, the Cauchy–Riemann equations put the real differential in this form. Dividing its remainder by h then gives the complex derivative. ∎

The existence of partial derivatives is not enough. The proposition assumes real differentiability. For a function with continuous first partial derivatives, real differentiability follows from real calculus, so the equations suffice.

The determinant of this matrix is $A^2 + B^2 = |f'(a)|^2$. If $f'(a) \neq 0$, multiplication by $f'(a)$ scales lengths by $|f'(a)|$ and rotates every vector by the same angle. It therefore preserves oriented angles between nonzero tangent vectors. This is the meaning of *conformal at a* used here. For $f(z) = z^2$ and $a \neq 0$, the local length scale is $2\,|a|$. At $a = 0$ the differential is zero, so this conformality conclusion does not apply.

9.2 Power series and the logarithm

A convergent power series is holomorphic in its disk. Lecture 10 proves that every holomorphic function is locally given by a power series.

Consider a power series with radius of convergence $R > 0$:

$$f(z) = \sum_{n=0}^{\infty} a_n (z-a)^n, \qquad |z-a| < R.$$

By Theorem 1.5, this series has a complex derivative in its disk, and both the series and its derivative series converge uniformly on every concentric closed disk $|z-a| \le r < R$. Thus the sum is holomorphic, with

$$f'(z) = \sum_{n=1}^{\infty} n a_n (z-a)^{n-1}.$$

In particular, define the exponential and the trigonometric functions by

$$e^z = \sum_{n=0}^{\infty} \frac{z^n}{n!}, \qquad \cos z = \frac{e^{\mathrm{i}z} + e^{-\mathrm{i}z}}{2}, \qquad \sin z = \frac{e^{\mathrm{i}z} - e^{-\mathrm{i}z}}{2\mathrm{i}}.$$

The exponential series has infinite radius of convergence. Termwise differentiation gives $(e^z)' = e^z$. The product of its absolutely convergent series gives $e^{z+w} = e^z e^w$. On the imaginary axis, the even and odd terms give the usual real cosine and sine. The functional equation

therefore gives $e^{x+\mathrm{i}y} = e^x(\cos y + \mathrm{i}\sin y)$. The real chain rule and Proposition 9.2 give the complex chain rule. It gives $(\sin z)' = \cos z$ and $(\cos z)' = -\sin z$.

The principal logarithm requires a domain. On the slit plane $U = \mathbb{C} \setminus (-\infty, 0]$, define

$$\log z = \ln|z| + \mathrm{i}\arg z,$$
$$-\pi < \arg z < \pi.$$

The symbol ln is reserved for a positive real argument. The principal value at a negative real number uses $\arg z = \pi$, but that ray is excluded from the holomorphic domain U. The exponential maps the strip $\{w : -\pi < \operatorname{Im} w < \pi\}$ bijectively onto U: its modulus is $e^{\operatorname{Re} w}$ and its argument in $(-\pi, \pi)$ is $\operatorname{Im} w$, so both real coordinates of w are determined uniquely. The displayed logarithm is its inverse.

To differentiate it without assuming a complex inverse theorem, use local polar coordinates. Near any point of U, real calculus gives

$$\mathrm{d}\ln|z| = \frac{x\,\mathrm{d}x + y\,\mathrm{d}y}{x^2 + y^2},$$
$$\mathrm{d}\arg z = \frac{-y\,\mathrm{d}x + x\,\mathrm{d}y}{x^2 + y^2}.$$

These smooth local expressions satisfy Cauchy–Riemann. By Proposition 9.2, the logarithm is holomorphic and

$$(\log z)' = \frac{x - \mathrm{i}y}{x^2 + y^2} = \frac{1}{z}, \qquad z \in U.$$

The choice of the slit is part of this statement.

9.3 Contour integrals and primitives

A *curve* in Ω will mean a continuous, piecewise C^1 map $\gamma : [a, b] \to \Omega$. Its length and the integral of a continuous function $f : \Omega \to \mathbb{C}$ along it are

$$L(\gamma) = \int_a^b |\gamma'(t)|\,\mathrm{d}t,$$
$$\int_\gamma f(z)\,\mathrm{d}z = \int_a^b f(\gamma(t))\gamma'(t)\,\mathrm{d}t.$$

Integrals across the finitely many smooth pieces are added. Reversing a curve changes the sign of its integral. Joining successive curves adds their integrals. Both statements follow by changing the real parameter. The triangle inequality gives the *ML-estimate*:

$$\left|\int_\gamma f(z)\,\mathrm{d}z\right| \le \sup_{z\in\gamma([a,b])} |f(z)|\; L(\gamma). \tag{9.1}$$

A *primitive* of f on Ω is a holomorphic function $F : \Omega \to \mathbb{C}$ with $F' = f$. Along a smooth piece of γ, the chain rule gives $(F \circ \gamma)'(t) = f(\gamma(t))\gamma'(t)$. The real fundamental theorem of calculus therefore gives

$$\int_\gamma f(z)\,\mathrm{d}z = F(\gamma(b)) - F(\gamma(a)). \tag{9.2}$$

In particular, the integral is zero when the curve is *closed*, meaning $\gamma(a) = \gamma(b)$. For $f(z) = z^2$, the primitive $F(z) = z^3/3$ gives a zero integral around every closed curve in $\mathbb{C}$.

A hole in the domain can prevent a primitive. Parametrise the unit circle counter-clockwise by $\gamma(t) = e^{\mathrm{i}t}$ for $0 \le t \le 2\pi$. Then

$$\oint_{|z|=1} \frac{\mathrm{d}z}{z} = \int_0^{2\pi} \frac{\mathrm{i}e^{\mathrm{i}t}}{e^{\mathrm{i}t}}\,\mathrm{d}t = 2\pi\mathrm{i}. \tag{9.3}$$

Hence $1/z$ has no primitive on $\mathbb{C} \setminus \{0\}$, although it is holomorphic there. Its primitive $\log z$ on the slit plane cannot be used on the whole circle. The same integral rules out a primitive on an annulus containing that circle. Optional comparison with Theoretical Mechanics, Lecture 8: the angular differential is the same local differential as $\mathrm{d}\arg z$ above, $\mathrm{d}\theta = (-y\,\mathrm{d}x + x\,\mathrm{d}y)/(x^2 + y^2)$. Its integral around this circle is 2π, which is the imaginary part of Equation (9.3).

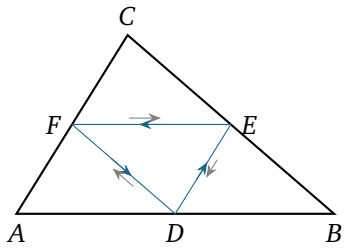
9.4 Goursat's theorem

For a filled triangle T, the symbol ∂T denotes its boundary traversed counter-clockwise. The whole filled triangle must lie in the open set Ω.

Theorem 9.3 (Goursat). *Let f be holomorphic on an open set $\Omega \subset \mathbb{C}$. If the filled triangle T is contained in Ω, then*

$$\oint_{\partial T} f(z)\,\mathrm{d}z = 0.$$

Joining the side midpoints divides the triangle into four smaller triangles.



The four small triangles carry the same counter-clockwise orientation. Each shared edge contributes twice with opposite signs.

Proof. We construct nested triangles. Write $I(T)$ for the boundary integral, d for the diameter of T and p for its perimeter. Joining the midpoints divides T into four triangles of half the diameter and half the perimeter. Their counter-clockwise boundary integrals add to $I(T)$, since each interior edge is traversed in opposite directions and the remaining half-edges make up ∂T. At least one subtriangle T_1 therefore satisfies $|I(T_1)| \ge |I(T)|/4$. Repeat this choice inside T_1. After n steps,

$$\begin{gathered} T_{n+1} \subset T_n, \\ |I(T_n)| \ge 4^{-n}\,|I(T)|\,, \\ d_n = 2^{-n} d, \quad p_n = 2^{-n} p. \end{gathered}$$

The triangles are compact, nested and nonempty. Their diameters tend to zero, so they have a unique common point $a \in T \subset \Omega$.

Complex differentiability at a gives

$$\begin{aligned} f(z) &= f(a) + f'(a)(z-a) \\ &\quad + \varepsilon(z)(z-a), \\ \varepsilon(z) &\longrightarrow 0 \quad (z \to a). \end{aligned}$$

Set $\varepsilon(a) = 0$ and let $\eta_n = \sup_{z \in T_n} |\varepsilon(z)|$. The remainder limit and $|z-a| \le d_n$ give $\eta_n \to 0$. The constant and linear terms have polynomial primitives, so their boundary integrals vanish by Equation (9.2). By the ML-estimate,

$$|I(T_n)| \le \eta_n d_n p_n = \eta_n 4^{-n} dp.$$

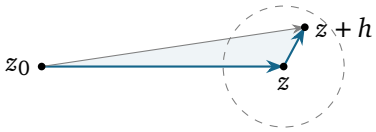
Combining the lower and upper bounds gives $|I(T)| \le \eta_n dp$ for every n. Letting $n \to \infty$ proves $I(T) = 0$. ∎

The proof assumes no continuity of f'. This matters because the next lecture will derive further regularity from Cauchy's formula.

9.5 A primitive on a star-shaped domain

A *domain* is a nonempty connected open subset of $\mathbb{C}$. A domain Ω is *star-shaped about* $z_0 \in \Omega$ if the whole segment $[z_0, z]$ lies in Ω for every $z \in \Omega$. Every disk is star-shaped about its centre.

The primitive construction compares the segments from z_0 to two nearby points. The disk at z supplies the short third side of the triangle.



The shaded triangle lies in the domain because every segment from z_0 to a point of the short side lies in the domain.

Theorem 9.4 (Cauchy's theorem on a star-shaped domain)**.** *Let $\Omega \subset \mathbb{C}$ be a domain star-shaped about z_0. Every holomorphic function $f : \Omega \to \mathbb{C}$ has a primitive on Ω. Consequently,*

$$\oint_\gamma f(z)\,\mathrm{d}z = 0$$

for every closed, piecewise C^1 curve γ in Ω.

Proof. Define a function by integrating along straight segments:

$$F(z) = \int_{[z_0, z]} f(\zeta)\,\mathrm{d}\zeta, \qquad z \in \Omega.$$

Fix $z \in \Omega$ and choose a disk about z contained in Ω. For sufficiently small h, the segment $[z, z+h]$ lies in this disk. The filled triangle with vertices $z_0, z, z+h$ is also in Ω: every one of its points lies on a segment from z_0 to a point of $[z, z+h]$. Goursat's theorem, with the boundary followed as $z_0 \to z \to z+h \to z_0$, gives zero in either orientation. Thus

$$0 = \int_{[z_0, z]} f(\zeta)\,\mathrm{d}\zeta + \int_{[z, z+h]} f(\zeta)\,\mathrm{d}\zeta - \int_{[z_0, z+h]} f(\zeta)\,\mathrm{d}\zeta.$$

Using the definition of F and parametrising the short segment gives

$$F(z+h) - F(z) = h \int_0^1 f(z+th)\,\mathrm{d}t.$$

If the triangle is degenerate, the same equality follows by cancellation of oriented line segments. Dividing by h gives

$$\left| \frac{F(z+h) - F(z)}{h} - f(z) \right| \leq \sup_{0 \leq t \leq 1} |f(z+th) - f(z)|.$$

The bound tends to zero by the continuity of f at z, which follows from its complex differentiability. Thus $F' = f$ on Ω. The assertion about closed curves follows from Equation (9.2). ∎

The construction gives $F(z_0) = 0$. For example, if $f(z) = z^2$ and $z_0 = 0$, integration along $\zeta = tz$ gives $F(z) = z^3 \int_0^1 t^2\,\mathrm{d}t = z^3/3$. For $f(z) = 1/z$, the slit plane of § 9.2 is star-shaped about 1: a segment from 1 to a nonreal point stays off the real axis except at 1, and a segment to a positive real point stays positive. The same construction gives $\log z$, since both primitives vanish at 1. Their difference is constant: its derivative vanishes, so its restriction to each segment from 1 is constant by real calculus.

9.6 Domains used in later contour arguments

The next contour arguments need the following domain checks. A set is *convex* if it contains the segment joining any two of its points. An open half-disk, for example $\{z : |z| < R,\ \operatorname{Im} z > 0\}$, is convex as the intersection of a disk and a half-plane. It is therefore star-shaped about each of its points. The theorem applies to closed curves inside this open half-disk. Boundary contours need a further argument, given in Lecture 11 where they are used to evaluate real integrals.

The keyhole domain $V = \mathbb{C} \setminus [0, \infty)$ is star-shaped about -1. Indeed, a point of $[-1, z]$ has the form $-(1-t) + tz$, where $0 \leq t \leq 1$. If $\operatorname{Im} z \neq 0$, the imaginary part is nonzero for $t > 0$. If $z \in V$ is real, it is negative, and the whole segment is negative. No segment meets the excluded ray. Lecture 11 uses this slit plane for integrals with a branch cut along the positive ray.

Lecture 11 uses annuli for Laurent expansions, which are series with positive and negative powers. An annulus $r < |z - a| < R$ is not star-shaped. For any candidate centre z_0, the reflected point $2a - z_0$ is also in the annulus. Their segment passes through the excluded point a. We use narrow annular sectors instead. After translating and rotating, consider

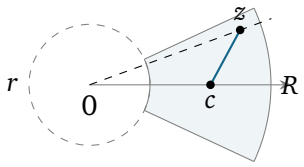
$$S = \{z : r < |z| < R,\ |\arg z| < \alpha\},$$
$$0 < \alpha < \frac{\pi}{2}, \qquad \cos\alpha > r/R.$$

Choose $c \in (r, R)$ such that $c\cos\alpha > r$. We show that S is star-shaped about the real point c. For $z = qe^{\mathrm{i}\theta} \in S$, write a point of the segment as $w = (1-t)c + tz$, where $0 \leq t \leq 1$. The point stays in the angular wedge, which is convex. Its outer-radius bound follows from $|w| \leq (1-t)c + tq < R$. For the inner bound, project onto the radial direction of z:

$$\operatorname{Re}(we^{-\mathrm{i}\theta})$$
$$= (1-t)c\cos\theta + tq > r.$$

Hence $|w| > r$, so the segment stays in S. The condition $\cos\alpha > r/R$ ensures that such a c exists.

The sector and a segment from its star centre are shown below.



The shaded sector has star centre c. The segment from c to z stays between the two circles. Finitely many overlapping sectors of sufficiently small angular width therefore cover an annulus. Cauchy's theorem applies separately on each sector. It does not assert that a holomorphic function on the entire annulus has zero integral around its central hole.

Summary.

- A complex derivative is a real differential given by multiplication by one complex number.
- Power series are holomorphic. The principal logarithm is holomorphic on its specified slit plane and has derivative $1/z$ there.
- Goursat's theorem gives zero boundary integrals for filled triangles contained in an open set on which the function is holomorphic.
- On a star-shaped domain, every holomorphic function has a primitive and every closed-curve integral is zero.

Lecture 10 uses these results to recover a function from its boundary values on a circle.

References

E. M. Stein and R. Shakarchi, *Complex Analysis*, Princeton University Press, 2003, Chapter 1, §§2–3, and Chapter 2, §§1–2 (Goursat and primitives, pp. 34–39).